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· Case Study - Final Year Dissertation

The Impact of Cosmetic Microtransactions on Player Engagement & Retention in Free-to-Play Games

BSc Computer Games Technology · University of West London · 2023 ·

PROBLEM

Free-to-play (F2P) games live or die on player retention. Functional microtransactions (pay-to-win boosts, time-savers) are a proven revenue lever but create fairness friction that alienates players. Cosmetic customisation sits in a different space: it generates revenue without altering gameplay balance but its actual effect on engagement and retention had not been tested with primary data in an academic context.

Research question: Do cosmetic customisation systems measurably influence player engagement and retention in free-to-play games?

WHY CLASH OF CLANS & CS:GO?

Clash of Clans

- F2P from day one (2012); cosmetic & time-saving microtransactions coexist
- Over £400 million revenue in 2022 despite being 10+ years old
- Provides insight into how cosmetics sustain engagement long-term

Counter-Strike: Global Offensive

- Paid game that transitioned to F2P in 2018, unlocking cosmetic sales
- Player base surged after transition; peak of 1.8 million concurrent players
- Pure cosmetics model (no pay-to-win) ideal comparison case

HYPOTHESIS

Cosmetic customisation features will increase player engagement (measured as playtime and interaction frequency) even in a minimal gameplay context, because personalisation drives a psychological sense of ownership and investment in the experience.

PROTOTYPE — SNAKE WITH COSMETIC DLC

A retro Snake game was built in Unity (C#) to generate primary data. Two cosmetic buttons Change Map Colour and Change Snake Colour were placed in the HUD, simulating a lightweight cosmetic system without any monetary gate. All button interactions were manually recorded during live screen-share sessions.

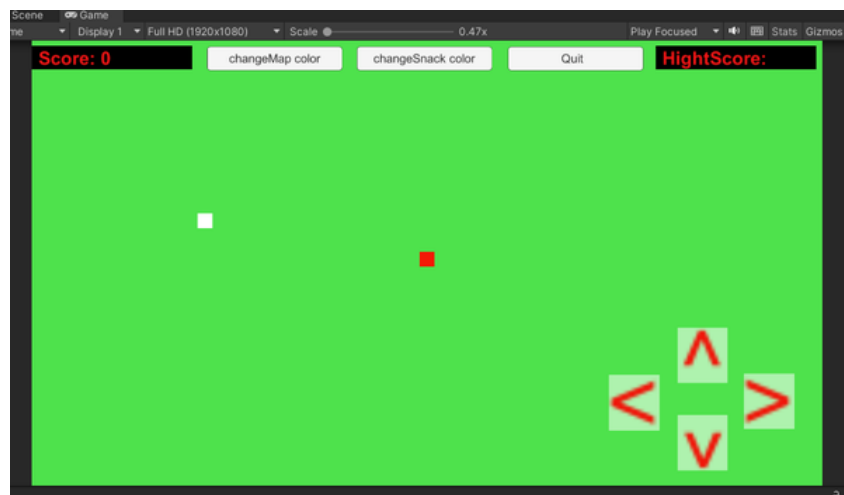


Figure 1 — The Unity Snake prototype. Cosmetic buttons visible at the top of the HUD.

METHOD

20 Participants	Discord Delivery	2 metrics Playtime + DLC presses	8 questions Post-play survey
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Data Collection

- Participants recruited via Discord gaming communities
- Each participant screen-shared live gameplay so button interactions could be manually logged in real time
- Dependent variables: total playtime (minutes) and number of DLC button presses
- Post-play: 8-question survey (Likert scales + open-ended)

Analysis

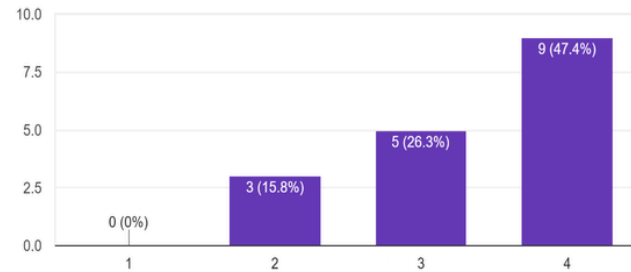
- Descriptive statistics for playtime and button-press frequency
- Pearson correlation coefficient between playtime and DLC interactions
- Survey responses: frequency counts (closed) and thematic coding (open-ended)
- Secondary data CoC & CS:GO revenue and playerbase provided industry context

RESULTS

1.83 min Avg. playtime	21.7 Avg. DLC presses	r = 0.74 Playtime ↔ Presses	70% Said cosmetics enhanced experience	75% Said cosmetics added value
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On a scale of 1-5 (1 being "Not at all" and 5 being "extremely"), how enjoyable did color-changing features in the game?

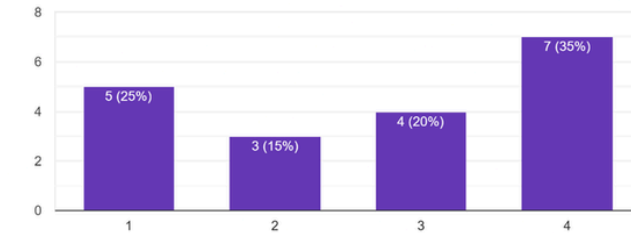
19 responses



Q1 — Enjoyment of colour-changing features (1–5 scale, n=19). 73% rated 4 or 5.

One a scale of 1-5, how likely are you to replay the game because of the color-chan being least likely - 5 being more likely)

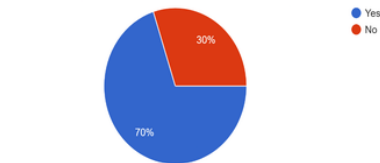
20 responses



Q4 — Replay likelihood due to cosmetics (1–5). 35% scored 4.

Did the ability to change the color of the snake and the background enhance your gaming experience?

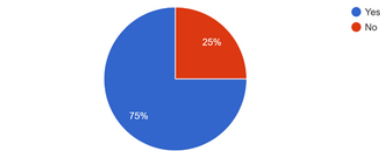
20 responses



Q2 — Did cosmetics enhance your experience? 70% Yes (n=20).

Do you feel the colour-changing features added value to the game?

20 responses



Q5 — Did cosmetics add value to the game? 75% Yes (n=20).

KEY FINDINGS

01 Cosmetics engage early Button interactions peaked in the first two minutes, cosmetics create an initial hook, but novelty fades without deeper game mechanics to sustain it.	02 Playtime ↔ Engagement link r = 0.74 correlation between playtime and DLC presses: players who stayed longer interacted with cosmetics more or vice versa.	03 Perceived value is real 75% felt cosmetics added value; 70% said they enhanced the experience despite the game being a minimal Snake clone.	04 Personalisation matters Qualitative responses cited 'owning my style', visual distinction, and reduced monotony consistent with Böffel (2022) on avatar attachment.
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WHAT I'D IMPROVE IN THE NEXT ITERATION

Study Design

- Control vs. experimental groups run identical sessions with and without cosmetics to isolate their effect on playtime
- Larger, more diverse sample 20 participants limits statistical power; 100+ across age, gender, and gaming frequency would strengthen claims
- Automated logging replace manual Discord observation with in-game telemetry to eliminate observer bias

Prototype Depth

- Tie cosmetics to progression unlock colours via achievements to better mirror real F2P design and test earned vs. freely available motivation
- More cosmetic variety textures, patterns, animated skins; survey respondents consistently requested richer options
- Longer sessions a more engaging core loop would allow meaningful retention curves rather than sub-2-minute average playtimes